



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

22-04-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	C	D	D	C	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A	B,C	A,B,C,D	A,B,C	A,C,D	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	7.00	5.00	5.00	1.00	0.83 to 0.85	4.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	B	C	D	D	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,D	A,B,C,D	B,C	B,D	B,C,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	270.00	2.00	0.00	5.50	3.00	13.00

PART-3 : MATHEMATICS

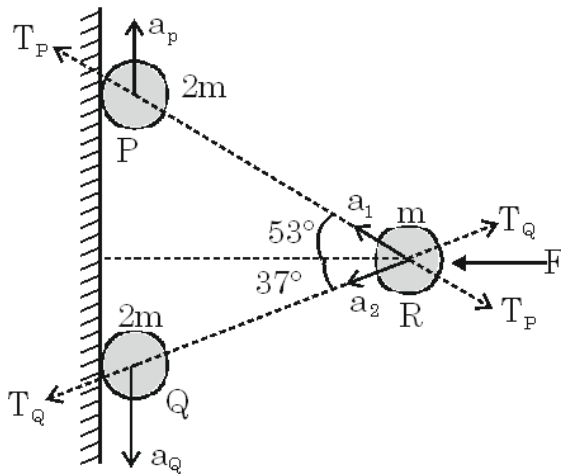
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	A	A	B	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,B,C	A,B,D	A,B,D	A,B,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	18.00	8.00	1.00	4.00	155.00	2.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)



Let component of acceleration of disc R are

a_1 and a_2

$$a_1 = a_P \cos 37^\circ = \frac{4a_P}{5} \quad \dots(1)$$

$$a_2 = a_Q \cos 53^\circ = \frac{3a_Q}{5} \quad \dots(2)$$

for P

$$T_P \cos 37^\circ = 2ma_P$$

$$\frac{4T_P}{5} = 2ma_P \quad \dots(3)$$

For Q

$$T_Q \cos 53^\circ = 2ma_Q$$

$$\frac{3T_Q}{5} = 2ma_Q \quad \dots(4)$$

For R

$$F \cos 53^\circ - T_P = ma_1$$

$$\frac{3F}{5} - T_P = ma_1 \quad \dots(5)$$

$$\text{and } F \cos 37^\circ - T_Q = ma_2$$

$$\frac{4F}{5} - T_Q = ma_2$$

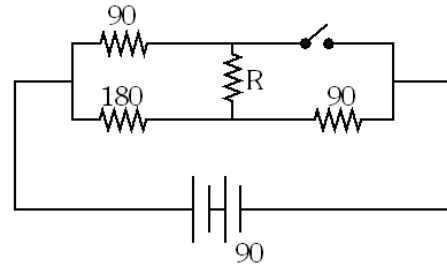
from (3) & (5)

$$\frac{3F}{5} - \frac{10}{4}ma_P = ma_1$$

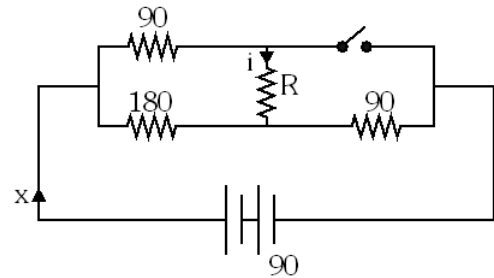
using (1) in above relation we get

$$a_P = \frac{2F}{11m}$$

2. Ans (C)



Let say current by battery is x and current in bulb is i



$$x = \frac{90}{\left(\frac{(90+R) \times 180}{270+R}\right) + 90} = \frac{270+R}{450+3R} \quad \dots(1)$$

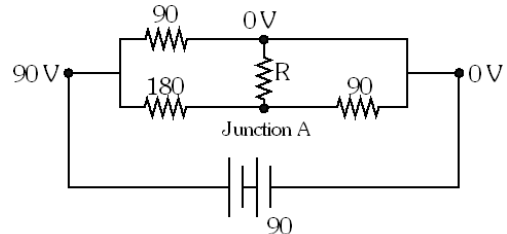
Since $(90+R)$ and 180 are in parallel

$$i(90+R) = (x-i)180$$

$$i = \frac{180x}{270+R} \quad \dots(2)$$

Solving eq. (1) & (2)

$$i = \frac{60}{150+R}$$



Let say potential of junction A is B and apply

Kirchhoff's junction law for A

$$\frac{90-V}{180} = \frac{V-0}{R} + \frac{V-0}{90}$$

$$V = \frac{\frac{1}{2}}{\frac{1}{180} + \frac{1}{90} + \frac{1}{R}}$$

$$\text{Current through bulb in first condition } i = \frac{60}{150+R}$$

$$\text{and current through bulb in second condition } i = \frac{V}{R}$$

$$\frac{60}{150+R} = \frac{V}{R} = \frac{\left(\frac{\frac{1}{2}}{\frac{1}{180} + \frac{1}{90} + \frac{1}{R}}\right)}{R}$$

Solving we will get $R = 30$

3. **Ans (D)**

$$V_{CM} = \frac{mv_0}{2m} = \frac{v_0}{2}$$

$$2mgh = \frac{2m\left(\frac{v_0}{2}\right)^2}{2}$$

$$h = \frac{v_0^2}{8g}$$

$$\Rightarrow K = \frac{mv_0}{2} - 2mgh = \frac{mv_0^2}{2} - \frac{mv_0^2}{4} = \frac{mv_0^2}{4}$$

4. **Ans (D)**

Initially,

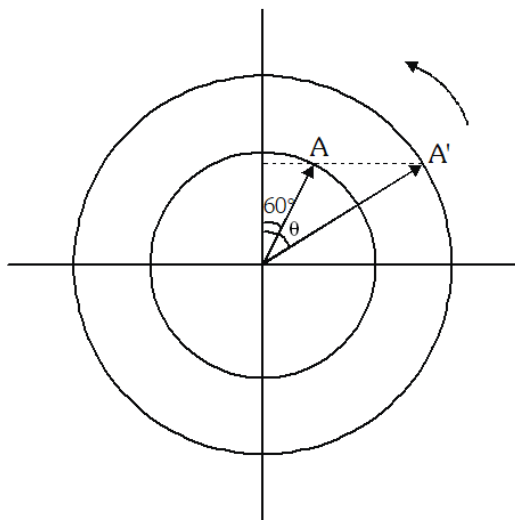
$$V_1 = \left| \frac{V}{L} \cdot \frac{L}{4} - V \right| = \frac{3V}{4}$$

Finally,

$$V_2 = \left| \frac{2V}{L} \cdot \frac{L}{4} - V \right| = \frac{V}{2}$$

$$\therefore \frac{V_1}{V_2} = \frac{3}{2}$$

5. **Ans (C)**



$$V = \omega \sqrt{A^2 - x^2}$$

$$V = \frac{\pi \sqrt{3}}{T} A$$

$$V' = 2V; \quad x = A/2$$

$$\frac{2\sqrt{3}\pi A}{T} = \frac{2\pi}{T} \sqrt{A'^2 - x^2}$$

$$A' = \frac{\sqrt{13}A}{2}$$

$$A' \cos \theta' = \frac{A}{2}; \quad \cos \theta' = \frac{1}{\sqrt{13}}$$

$$\theta' = \omega t$$

$$t = \frac{T}{2\pi} \cos^{-1} \left(\frac{1}{\sqrt{13}} \right)$$

6. **Ans (C)**

Write co-ordinates of bottom point on the rim at time t and find locus.

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**

From the given situation,

$$F - \alpha v^2 = mv \frac{dv}{dx}$$

$$\Rightarrow \frac{F - \alpha v^2}{F - \alpha v_0^2} = e^{-2\alpha x/m}$$

$$\therefore F = \alpha \left[\frac{v^2 - v_0^2 e^{-2\alpha x/m}}{1 - e^{-2\alpha x/m}} \right]$$

8. **Ans (B,C)**

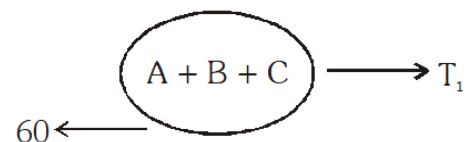
$$f_{01} = 50$$

$$f_{02} = 100$$

$$f_{03} = 60$$

There is a relative motion between C and D.

A + B + C] blocks move together.



$$T_1 - 60 = 30a_1 \quad \dots(1)$$

$$100 - T_1 = 10a_1 \quad \dots(2)$$

$$a_1 = 1 \text{ m/s}^2$$

$$E : 500 - T_2 = 50 a_2 \quad \dots(3)$$

$$D : T_2 - 60 = 10 a_2 \quad \dots(4)$$

$$a_2 = \frac{22}{3}$$

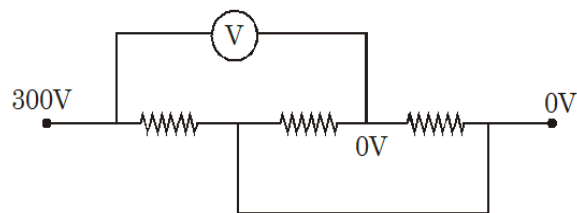
$$T_2 = \frac{400}{3} \text{ N}$$

9. Ans (A,B,C,D)

Ideal voltmeter ($R_V = \infty$)

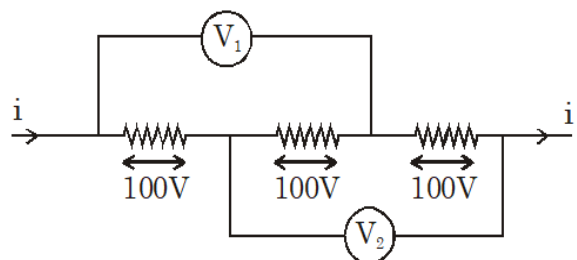
Ideal ammeter ($R_A = 0$)

(A)



Reading = 300V.

(B)

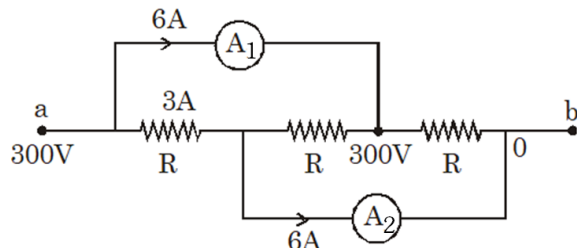


$$i = \frac{300}{300} = 1A$$

$$V_1 = 100 + 100 = 200V$$

$$V_2 = 100 + 100 = 200V$$

(C)

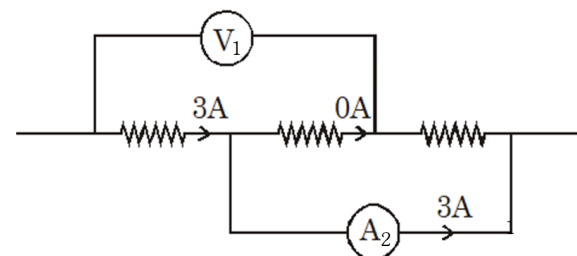


All resistance are in parallel combination.

$$i = \frac{300}{100/3} = 9A$$

Reading of ammeter is 6A

(D)



$$i = \frac{300}{100} = 3A$$

10. Ans (A,B,C)

$$q = 2e$$

$$n = \left(\frac{i}{q} \right) = 6.25 \times 10^{11} \text{ per sec.}$$

in 4 sec. = $4 \times n = 2.5 \times 10^{12}$ particles spikes the surface per sec.

$$E = 83.5 \text{ KeV}$$

$$\frac{1}{2}mu^2 = E$$

$$u = 2 \times 10^6 \text{ m/s}$$

$$E = q(V)$$

$$V = 41.75 \text{ KV}$$

$$P = Vi = 8.35 \times 10^{-3} \text{ W}$$

11. Ans (A,C,D)

The system can be assumed to be combination of a uniform sphere with four small sphere of -ve mass.

12. Ans (A,C,D)

$$\frac{n\lambda}{2} = L \Rightarrow \lambda = \frac{2L}{n} \Rightarrow f = \frac{v}{\lambda} = \frac{n}{2L} \sqrt{\frac{T}{\mu}}$$

(A) for same n , T & μ , if L is doubled, f is halved.

(C) for same f , L & μ , $n\sqrt{T} = \text{constant}$

$$(n+1)\sqrt{T'} = n\sqrt{T} \Rightarrow T' = \frac{n^2}{(n+1)^2} \cdot T$$

$$(D) f = \frac{n}{2L} \sqrt{\frac{T}{\mu}}$$

$$3f = \frac{(2n)}{2\left(\frac{L}{2}\right)} \sqrt{\frac{T'}{\mu}}$$

$$T' = \frac{9T}{16}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (7.00)**

$$\frac{Pr^3}{R^3} = kA \frac{dT}{dr} = k \cdot 4\pi r^2 \frac{dT}{dr}$$

$$\frac{P}{R^3} \int r dr = 4\pi k \int dT$$

$$\Rightarrow \frac{PR^2}{2R^3} = 4\pi k (T_C - T_S)$$

$$\Rightarrow \frac{P}{2R} = 4\pi k \Delta T$$

$$\Rightarrow P = 8\pi k R \Delta T = eA\sigma T_S^4$$

$$\Rightarrow \Delta T = \frac{4\pi R^2 \sigma \cdot T_S^4}{8\pi k R}$$

$$\Rightarrow \Delta T = \frac{\sigma R^2 \cdot T_S^4}{2kR} = \frac{\sigma R T_S^4}{2k}$$

$$\Rightarrow m = 1, n = 4, \ell = 2$$

$$\Rightarrow m + n + \ell = 7$$

2. **Ans (5.00)**

$$\beta = \frac{\lambda P}{d} = \frac{5000 \times 10^{-10} \times 10}{\sqrt{2} \times 10^{-3}} = \frac{5}{\sqrt{2}} \times 10^{-3} \text{ m}$$

Now, since the screen can be assumed to be

inclined at 45° w.r.t. slits, hence fringe width

on the screen

$$\frac{\beta}{\cos \theta} = \frac{5}{\sqrt{2}} \times 10^{-3} \times \frac{1}{\cos 45^\circ} = 5 \times 10^{-3} \text{ m}$$

3. **Ans (5.00)**

F_{net} due to radiation = $\frac{I\pi R^2}{C}$ in the direction of radiation.

$$\therefore F_H = \frac{I\pi R^2}{C} \cos 60^\circ = \frac{I\pi R^2}{2C}$$

This force is balanced by spring force.

$$\therefore Kx = \frac{I\pi R^2}{2C} \Rightarrow x = \frac{I\pi R^2}{2KC} = 5$$

4. **Ans (1.00)**

Natural length = 3.45 cm

$$K(0.04) = 10$$

$$K = 250 \text{ N/cm}$$

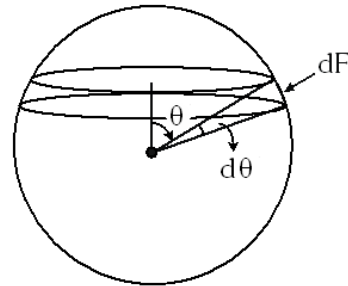
$$\text{Now } K' = \frac{250}{1.15} (3.45) = 750$$

$$750 \frac{(5-x)}{100} = 30$$

$$5-x = \frac{3000}{750} = 4$$

$$x = 1$$

5. **Ans (0.83 to 0.85)**



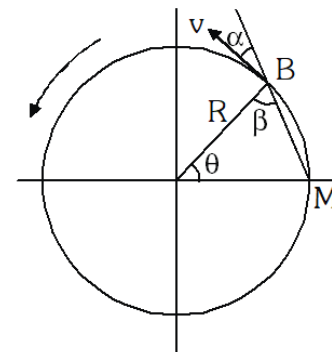
$$dF = (Rd\theta) (2\pi R \sin \theta) \left(\frac{R}{2} - R \cos \theta \right) \times \rho g$$

$$F_y = \int_{\pi/3}^{\pi} dF \cos \theta = \frac{9}{8} \pi R^3 \rho g \quad (\text{total upward buoyancy force})$$

at equilibrium buoyancy force = weight of the floating object

6. **Ans (4.00)**

Let the position of Buzzer at time t be B and that of the mic be M.



$$V = \omega R; \quad \theta = \omega t; \quad \beta = \frac{\pi - \theta}{2}$$

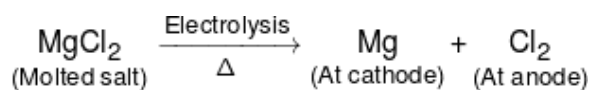
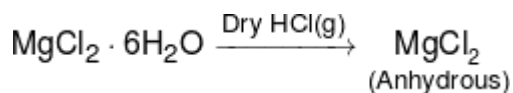
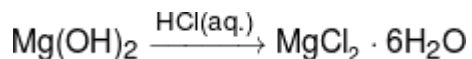
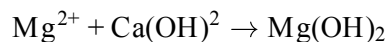
$$\alpha = \frac{\pi}{2} - \beta = \frac{\theta}{2} = \frac{\omega t}{2}$$

$$\therefore f = f_0 \left[\frac{V_0}{V_0 + V \cos \alpha} \right] = f_0 \left[\frac{V_0}{V_0 + \omega R \cos \left(\frac{\omega t}{2} \right)} \right]$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)



3. Ans (C)

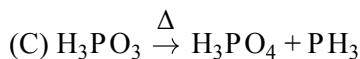
Theory based.

Enzymes catalyse reaction at around pH 5–7.

4. Ans (D)

(A) $\text{NO}_2(\text{g})$ will be produced.

(B) Bond length are different.



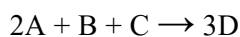
(D) Theory based.

5. Ans (D)

% of Bromine

$$\begin{aligned}
 &= \frac{80 \times \text{Weight of AgBr}}{188 \times \text{Weight of organic compound}} \times 100\% \\
 &= \frac{80 \times 0.12}{188 \times 0.15} \times 100 = 34.04\%
 \end{aligned}$$

6. Ans (B)



$$\text{Rate} = \text{K}[\text{A}]^x[\text{B}]^y[\text{C}]^z$$

for exp. (2) and (4)

$$\frac{r_1}{r_2} = 1 = \left(\frac{1}{2}\right)^z \Rightarrow z = 0$$

for exp. (1) and (2)

$$\frac{r_1}{r_2} = 1 = \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^y \Rightarrow 2^{-(x+y)} = 1$$

$$\Rightarrow x + y = 0 \quad \dots\text{(i)}$$

for exp. (1) and (3)

$$\frac{r_1}{r_3} = \frac{10^{-3}}{5 \times 10^{-4}} = \left(\frac{1}{4}\right)^y \left(\frac{1}{2}\right)^z \left(\frac{1}{2}\right)^x \Rightarrow 2 = \left(\frac{1}{4}\right)^y \times \left(\frac{1}{2}\right)^x$$

$$2 = \left(\frac{1}{2}\right)^{2y+x}$$

$$2y + x = -1 \quad \dots\text{(ii)}$$

Solving (i) and (ii)

$$\Rightarrow x = 1; y = -1$$

$\therefore$ Order with respect to A = 1

Order with respect to B = -1

Overall order = 1 - 1 + 0 = 0

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (A,B,D)**

$$U = nC_V T = aV^\alpha \text{ so } TV^{-\alpha} = a$$

$$\because \text{for polytropic process } TV^{n-1} = \text{const. so } n = 1 - \alpha$$

Molar heat capacity of

$$C_V - \frac{R}{n-1} \Rightarrow \frac{R}{\gamma-1} - \frac{R}{1-\alpha-1} = \left(\frac{R}{\alpha} - \frac{R}{1-\gamma} \right)$$

$$\because q = C\Delta T$$

$$= \left(C_V + \frac{R}{\alpha} \right) \Delta T \left(\because C = C_V + \frac{R}{\alpha} \right)$$

$$= C_V \Delta T + \frac{R\Delta T}{\alpha} = \Delta U + \frac{R}{\alpha} \Delta T = \Delta U + \frac{C_V \cdot (\gamma-1)}{\alpha} \Delta T$$

$$\Rightarrow \Delta U + \Delta U \frac{(\gamma-1)}{\alpha} \Rightarrow \Delta U \left(1 + \frac{\gamma-1}{\alpha} \right)$$

8. **Ans (A,B,C,D)**

(A) $O > N > B > C$.

(B) $O > C > Li > B > N$.

(C) 6th period & d-block.

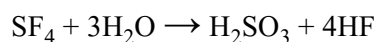
(D) $B > Tl > Ga > Al > In$.

10. **Ans (B,D)**

Above T_B , slope of Z v/s $P > 0$ (at low pressure)

Also valid for H_2 and He above critical temperature.

11. **Ans (B,C,D)**



PART-2 : CHEMISTRY

SECTION-II

3. **Ans (0.00)**

4 optically inactive compound will be obtained.

4. **Ans (5.50)**

No. of tetrahedral void = 4

$$\text{No. of octahedral void} = 2 \times \frac{1}{4} + 1$$

$$\text{Total effective no. of void} = \frac{11}{2}$$

5. **Ans (3.00)**

(a) $\Rightarrow$ No POS

(b) $\Rightarrow$ POS + diamagnetic & $Pt^{2+} \Rightarrow 3d^8 4s^0$

(SFL)

(c) $\Rightarrow$ POS + diamagnetic & $Co^{3+} \Rightarrow t_g^6 e_g^0$

(d) $\Rightarrow$ POS + diamagnetic & $Ni^0 \Rightarrow t_g^6 e_g^4$

(e) Paramagnetic

(f) Show optical activity

6. **Ans (13.00)**

$P = 4$, $Q = 3$ and $R = 6$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**

$$\tan^{-1} \left(\frac{1 - \sin 5}{\cos 5} \right)$$

$$= \tan^{-1} \left(\frac{1 - \cos \left(\frac{\pi}{2} - 5 \right)}{\sin \left(\frac{\pi}{2} - 5 \right)} \right)$$

$$= \tan^{-1} \left(\tan \left(\frac{\pi}{4} - \frac{5}{2} \right) \right) = \frac{5\pi}{4} - \frac{5}{2}$$

2. **Ans (A)**

$$\frac{1}{\sqrt{k} + \sqrt{k+1}} = \sqrt{k+1} - \sqrt{k}$$

$$\Rightarrow \sum_{k=1}^{80} (\sqrt{k+1} - \sqrt{k}) < \sum_{k=1}^{80} \frac{1}{2\sqrt{k}}$$

$$\Rightarrow \sum_{k=1}^{80} \frac{1}{\sqrt{k}} > 16$$

similarly

$$\sum_{k=2}^{80} (\sqrt{k} - \sqrt{k-1}) > \sum_{k=1}^{80} \frac{1}{2\sqrt{k}}$$

$$\Rightarrow S = (1 + 2\sqrt{80} - 2) < 17$$

3. **Ans (A)**

$$y = \left| S + C + \frac{S}{C} + \frac{C}{S} + \frac{1}{C} + \frac{1}{S} \right|$$

$$= \left| S + C + \frac{1}{SC} + \frac{S+C}{SC} \right|$$

$$= \left| S + C + \frac{S+C+1}{SC} \right|$$

$$= \left| S + C + \frac{(S+C+1)}{(S+C)^2 - 1} \times 2 \right|$$

$$= \left| S + C + \frac{2}{S+C-1} \right|$$

$$= \left| S + C - 1 + \frac{2}{S+C-1} + 1 \right|$$

$$\text{Let } S + C - 1 = t$$

$$y = \left| t + \frac{2}{t} + 1 \right|, t \in [-\sqrt{2} - 1, \sqrt{2} - 1]$$

y is minimum when $t = -\sqrt{2}$

$$(f)_{\min} = 2\sqrt{2} - 1$$

4. **Ans (B)**

$$I = \int_0^{2\pi} \frac{|\sin \theta| d\theta}{1 + |\cos 2\theta|} = 4 \int_0^{\pi/2} \frac{|\sin \theta| d\theta}{1 + |\cos 2\theta|}$$

(apply Queen Theorem)

$$= 4 \int_0^{\pi/2} \frac{\sin \theta d\theta}{1 + |\cos 2\theta|} = 4 \int_0^{\pi/2} \frac{\cos \theta d\theta}{1 + |\cos 2\theta|}$$

$$\Rightarrow 2I = 4 \int_0^{\pi/2} \frac{\sin \theta + \cos \theta}{1 + |\cos 2\theta|} d\theta$$

$$= 8 \int_0^{\pi/4} \frac{4 \sin \theta + \cos \theta}{1 + \cos 2\theta} d\theta$$

$$= 8 \int_0^{\pi/4} \frac{\sin \theta + \cos \theta}{2\cos^2 \theta} d\theta$$

$$I = 2[\sec \theta + \ln(\sec \theta + \tan \theta)]_0^{\pi/4}$$

$$= 2[\sqrt{2} - 1 + \ln(\sqrt{2} + 1)]$$

5. **Ans (D)**

$$\operatorname{sgn}(x^2 - x + 1) = \operatorname{sgn}\left(\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}\right) = 1$$

$$\text{for } x \in [0, 3], x^2 - x + 1 \in \left[\frac{3}{4}, 7\right]$$

$$[x^2 - x + 1] \text{ is discontinuous where } x^2 - x + 1$$

$$= 1, 2, 3, 4, 5, 6, 7$$

$$\text{but } x^2 - x + 1 = 1 \Rightarrow x = 0, 1$$

$\therefore$ Discontinuous at 8 points.

6. Ans (C)

$$\text{Total ways} \Rightarrow \frac{4.6.5.3(\alpha+1)}{2}$$

$$\text{factors having HCF } 1 \Rightarrow 2^{5-1} = 16$$

$$\text{factors having HCF } 5 \Rightarrow 2^{4-1} \cdot 2 = 16$$

$$\text{factors having HCF } 7 \Rightarrow 2^{4-1} \cdot 1 = 8$$

$$\text{Probability} \Rightarrow \frac{40.2}{4.6.5.3(\alpha+1)} = \frac{1}{27}$$

$$\Rightarrow \alpha = 5$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,B,C)

Let $S(h, k)$ is focus of parabola then using

property of parabola

$$h^2 + (k-1)^2 = \left(\frac{1}{\sqrt{m^2+1}} \right)^2 \dots (1)$$

$$\text{Similarly } (h-1)^2 + k^2 = \left(\frac{m}{\sqrt{m^2+1}} \right)^2 \dots (2)$$

Using (1) and (2)

$$h^2 + (k-1)^2 + (h-1)^2 + k^2 = 1$$

$$\Rightarrow h = \frac{1}{2}, k = \frac{1}{2}$$

8. Ans (A,B,C)

$$I_n = \int_{-\pi}^{\pi} \left(\frac{2^x}{1+2^x} \right) \frac{\sin^2 \frac{nx}{2}}{\sin^2 \frac{x}{2}} dx$$

$$\Rightarrow 2I_n = \int_{-\pi}^{\pi} \frac{\sin^2 \frac{nx}{2}}{\sin^2 \frac{x}{2}} dx = 2 \int_0^{\pi} \left(\frac{\sin(nx/2)}{\sin(x/2)} \right)^2 dx$$

$$\Rightarrow I_n = \int_0^{\pi} \frac{\sin^2(nx/2)}{\sin^2 x/2} dx$$

$$\Rightarrow I_n - I_{n-1} = \int_0^{\pi} \frac{\sin^2(nx/2) - \sin^2((n-1)x/2)}{\sin^2 x/2} dx$$

$$\int_0^{\pi} \frac{\sin \left(nx - \frac{x}{2} \right)}{\sin \left(\frac{x}{2} \right)} dx = V_n (\text{let})$$

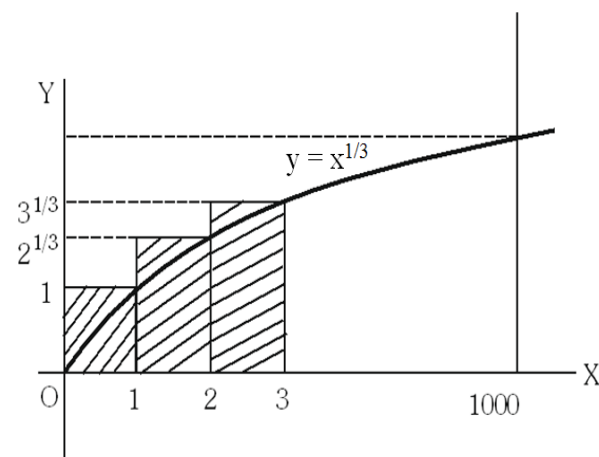
$$V_n - V_{n-1} = 0$$

$$\Rightarrow I_n - I_{n-1} = V_n - V_1 = \pi$$

Hence $I_1, I_2, I_3, \dots, I_n$ A.P.

$$I_1 = \pi, I_2 = 2\pi, \dots, I_n = n\pi$$

9. Ans (A,B,D)



S = sum of area of all shaded rectangles

$$S > \int_0^{1000} x^{1/3} dx \Rightarrow S > 7500$$

Parallely

$$(S - 10) < \int_0^{1000} x^{1/3} dx$$

$$\Rightarrow x < 7510$$

10. Ans (A,B,D)

$$xyz = 2^4 \times 3^3 \times 5^2 \quad \text{if } x, y, z$$

$${}^6C_2 \times {}^5C_2 \times {}^4C_2 = 900$$

if any 2 of them are negative

$$900 \times 3 = 2700$$

$$\text{Total solutions} = 3600$$

11. Ans (A,B,D)

$$f(f(x)) > f(2x - 7)$$

$$f'(x) = 6x^2 - 6x - 12 = 6(x - 2)(x + 1)$$

Case I

Case II

$$x \in (-1, 2)$$

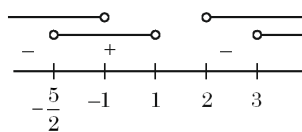
$$f(x) < 2x - 7$$

$$f(x) > 2x - 7$$

$$\Rightarrow 2x^3 - 3x^2 - 14x + 15 < 0$$

$$\Rightarrow (2x + 5)(x - 3)(x - 1) > 0$$

$$\Rightarrow (x - 1)[2x^2 - x - 15] < 0$$



$$\Rightarrow (2x + 5)(x - 3)(x - 1) < 0$$

$$x \in \left(-\frac{5}{2}, -1\right) \cup (3, \infty)$$

$$\Rightarrow x \in (1, 2)$$

$$x \in \left(-\frac{5}{2}, -1\right) \cup (1, 2) \cup (3, \infty)$$

12. Ans (B,C)

$$\text{Put } z = x + iy$$

$$\Rightarrow x^2 - y^2 + 2ixy - 2iy = 4i$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 0 \Rightarrow x = y \text{ or } -y \\ xy - y = 2 \end{cases}$$

$$\text{Case (1) if } x = y \text{ then } x^2 - x - 2 = 0$$

$$\Rightarrow x = -1, 2$$

$$\Rightarrow \begin{cases} z_1 = -1 - i \text{ (rejected)} \\ z_2 = 2 + 2i \end{cases}$$

$$\text{Case (2) if } x = -y, \text{ then } x^2 - x + 2 = 0$$

$$\Rightarrow x \in \phi$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (18.00)

$$A = \{-2, -1, 0, 1, 2\}$$

$$\vec{v}_1 \cdot \vec{v}_2 = 0$$

$$\Rightarrow a + b + c = 0$$

2. Ans (8.00)

Draw graph of both the functions.

3. Ans (1.00)

Coefficient of x^{-6} in

$$\left(1 - \frac{1}{x}\right) \left(1 - \frac{2}{x^2}\right) \dots \left(1 - \frac{14}{x^{14}}\right)$$

$$6 \rightarrow (0, 6), (1, 5), (2, 4), (1, 2, 3)$$

$$\text{So, required coefficient } -6 + 5 + 8 - 6 = 1$$

4. **Ans (4.00)**

$${}^{2n}C_n < \sum_{r=0}^{2n} {}^{2n}C_r = (2)^{2n}$$

$$\Rightarrow ({}^{2n}C_n)^{\frac{1}{n}} < 4$$

$$\text{also } ({}^{10}C_5)^{\frac{1}{5}} > 3$$

hence, $m = 4$

5. **Ans (155.00)**

$$x^2 - 5x - 2 = 0$$

$$S = \frac{\alpha^{2023} + (\alpha\beta)^3 \alpha^{2017} + \beta^{2023} + (\alpha\beta)^3 (\beta)^{2017}}{\alpha^{2020} + \beta^{2020}}$$

$$= \frac{\alpha^{2020}(\alpha^3 + \beta^3) + \beta^{2020}(\alpha^3 + \beta^3)}{\alpha^{2020} + \beta^{2020}}$$

$$= \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

$$= 125 - 3 \times (-2) (5)$$

$$= 125 + 30 = 155$$

6. **Ans (2.00)**

$$\text{adj}(AB) = |AB|^2 = (p-1)^2$$

$$|B^2 \text{adj} A| = |B^2| |A|^2 = |AB|^2$$

$$\Rightarrow (p-1)^2 = p^2 - 3 \Rightarrow p = 2$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,B,C,D	A,C	A,B,C	A,C	A,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	4.68	5.00	4.20	2.25	20.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	6	6	4	3	1

PART-2 : CHEMISTRY

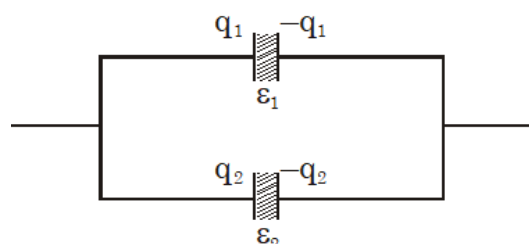
SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C,D	A,B	A,C	B,D	B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	32.93 to 33.00	51.20	14.00	3.47 or 3.48	1.40	12.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	4	9	2	5	2

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C	A,B,C,D	B,C	A,B,C	A,B,C	A,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.08	0.75	1.60	1.33	0.83	0.66 or 0.67
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	7	0	2	0	9

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (B,C)



as potential difference is same

$$\therefore \frac{q_1}{C_1} = \frac{q_2}{C_2}$$

$$C_1 = \frac{2\pi\epsilon_1 ba}{(b-a)}, C_2 = \frac{2\pi\epsilon_2 ab}{b-a}$$

$$\Delta V = \frac{q_1}{C_1} = \frac{q_2}{C_2} = \frac{Q(b-a)}{2\pi(\epsilon_1 + \epsilon_2)ab}$$

$$q_1 + q_2 = Q$$

$$q_1 = \frac{C_1}{C_1 + C_2} \times Q$$

$$q_1 = \frac{Q\epsilon_1}{\epsilon_1 + \epsilon_2}$$

$$q_2 = \frac{Q\epsilon_2}{\epsilon_1 + \epsilon_2}$$

$$\Delta V = \frac{q_1}{C_1} = \frac{\frac{Q\epsilon_1}{\epsilon_1 + \epsilon_2}}{\frac{2\pi\epsilon_1 ba}{b-a}}$$

$$\Delta V = \frac{Q(b-a)}{2\pi(\epsilon_1 + \epsilon_2)ba}$$

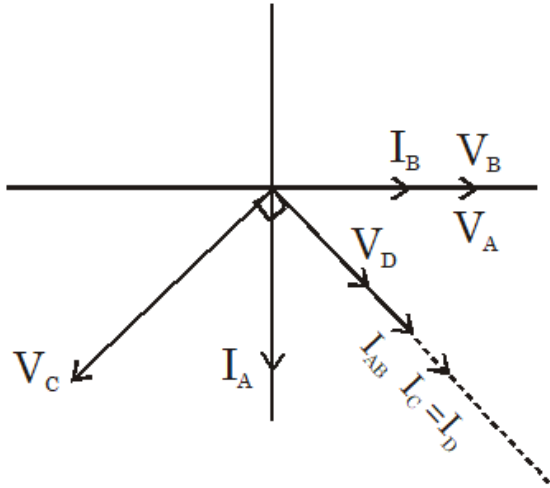
Using Gauss Law

$$E_1 \times 2\pi r^2 = \frac{q_1}{\epsilon_1}; E_2 \times 2\pi r^2 = \frac{q_2}{\epsilon_2}$$

$$E_1 = \frac{q_1}{\epsilon_1 2\pi r^2}$$

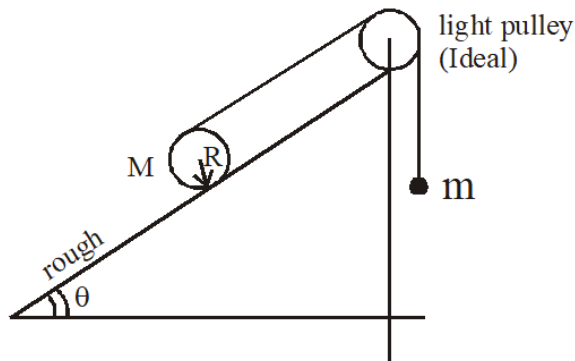
$$E_1 = \frac{Q}{(\epsilon_1 + \epsilon_2) 2\pi r^2} = E_2$$

2. Ans (A,B,C,D)



Phasor diagram of given circuit

3. Ans (A,C)



1. $T - mg = ma$

2. $Mg \sin \theta - T - f = M \frac{a}{2}$

3. $(f - T)R = \frac{MR^2}{2} \cdot \alpha$

4. $\frac{a}{2} = R \alpha$

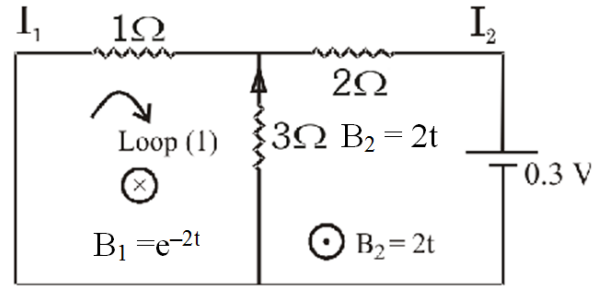
4. Ans (A,B,C)

$dQ = (4m)s(-dT_1) = (2m)(2s)dT_2$

On solving we get $\Delta T_f = \Delta T_i e^{\frac{-KA\ell}{2ms\ell}}$

Temperature of point P remains constant

5. Ans (A,C)



For loop 1 $\phi_1 = 0.2 \times e^{-2t}$

$E_1 = -\frac{d\phi_1}{dt} \bigg|_{t=0} = 0.2(-2)e^{-2t} \bigg|_{t=0} = -0.4 \text{ volt}$

$\Rightarrow \curvearrowright 0.4 \text{ V in Anticlockwise}$

$E_2 = -\frac{d}{dt}(0.2 \times 2t) = 0.4 \text{ in clockwise direction}$

Now in loop 1

$0.4 - I_1 - 3(I_1 + I_2) = 0$

$\Rightarrow 0.4 = 4I_1 + 3I_2 \quad \dots (1)$

in loop 2

$0.4 - 0.3 = 3(I_1 + I_2) + 2I_2$

$0.1 = 3I_1 + 5I_2 \quad \dots (2)$

by $3 \times (1) - 4 \times (2)$

$1.2 = 12I_1 + 9I_2$

$0.4 = 12I_1 + 20I_2$

$\underline{\quad \quad \quad}$

$0.8 = 0 = 11I_2$

$\Rightarrow I_2 = \frac{0.8}{11} = \frac{8}{110} \text{ A}$

now $I_1 = \frac{0.1 - 5I_2}{3} = \frac{\frac{1}{10} - 5\left(-\frac{8}{110}\right)}{3}$

$I_1 = \frac{\frac{1}{10} + \frac{40}{110}}{3}$

6. Ans (A,C)

Velocity of particle after time 't' is given by

$$v = gt$$

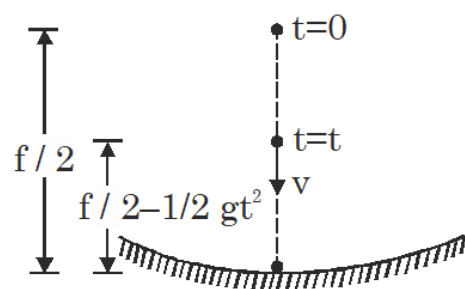
Distance travelled in time 't' is given by

$$S = \frac{1}{2}gt^2$$

Now, we know that

$$\vec{V}_{I/M} = -m^2 \vec{V}_{O/M}$$

Here, $\vec{V}_{O/M} = gt$



Also,

$$m = \frac{f}{f-u} = \frac{-f}{-f + \left(\frac{f}{2} - \frac{gt^2}{2}\right)} = \frac{2f}{f + gt^2}$$

V_1 = Velocity of image

$$= -\left(\frac{2f}{f + gt^2}\right)^2 \cdot gt$$

$$= \frac{4f^2 gt}{(f + gt^2)^2} \quad (- \text{ve sign indicates that image will move upward})$$

will move upward)

For maximum speed of image

$$\frac{dv_1}{dt} = 0$$

$$\Rightarrow t = \sqrt{\frac{f}{3g}}$$

$$v_{\max} = \frac{3}{4} \sqrt{3fg}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (8.00)

Change in momentum of 1 particle = $2 Mv_0 \cos \theta$

No. of particle striking per second per unit length

$$= (nv) R d\theta \cos \theta$$

$$f = \int_{-\pi/2}^{\pi/2} (nv R \cos \theta) d\theta [2mv \cos \theta] \cos \theta = 4nmv^2 R \int_0^{\pi/2} \cos^3 \theta d\theta$$

$$= \frac{4nmv^2 R}{4} \int_0^{\pi/2} (\cos 3\theta + 3 \cos \theta) d\theta$$

$$= nmv^2 R \left[\frac{\sin 3\theta}{3} + 3 \sin \theta \right] \Big|_0^{\pi/2} = \frac{8nmv^2 R}{3}$$

2. Ans (4.68)

$$\frac{dN}{dt} = q - \lambda N \quad (q = 10^8 \text{ nuclei/sec})$$

$$\Rightarrow \frac{dN}{q - \lambda N} = dt \Rightarrow \int_0^N \frac{dN}{q - \lambda N} = \int_0^t dt$$

$$\Rightarrow -\frac{1}{\lambda} \ln \left(\frac{q - \lambda N}{q} \right) = t$$

$$\Rightarrow t = \frac{t_{1/2}}{\ln 2} \cdot \ln \left(\frac{q}{q - A} \right)$$

$$\left(\because \lambda = \frac{\ln 2}{t_{1/2}} \right) \text{ and } A = \lambda N$$

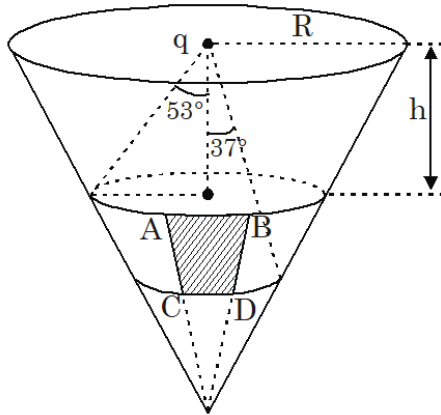
$$= \frac{t_{1/2}}{\ln 2} \cdot \ln \frac{10^8}{(10^8 - 7.5 \times 10^7)} = \frac{t_{1/2}}{\ln 2} \cdot \ln \left(\frac{10}{10 - 7.5} \right)$$

$$= 2t_{1/2} = 4.68 \text{ minute}$$

3. Ans (5.00)

Sol. $\frac{r}{h} = \tan 53^\circ$

$r = h \frac{4}{3}$



Flux through slant surface of cone containing

shaded section

$$\frac{q}{2\epsilon_0} [1 - \cos 53^\circ] - \frac{q}{2\epsilon_0} [1 - \cos 37^\circ]$$

$$= \frac{q}{2\epsilon_0} \left[\frac{4}{5} - \frac{3}{5} \right] = \frac{q}{2\epsilon_0} \frac{1}{5} = \frac{q}{10\epsilon_0}$$

$$\phi_{\text{through shaded section}} = \frac{\frac{q}{10\epsilon_0} \cdot \frac{h}{2}}{2\pi \cdot 4h/3}$$

$$\phi = \frac{3q}{160\pi\epsilon_0} = 5$$

4. Ans (4.20)

$$LC = \frac{\text{Pitch}}{\text{no. of circular divisions}} = \frac{0.75}{50} = 0.015 \text{ mm}$$

Negative zero error :

$$20 \text{ divisions} \times LC$$

$$\Rightarrow 20 \times 0.015 = (-0.3)$$

When metal is put

$$\text{Measured value} = 3.75 \text{ mm} + 10 \text{ divisions} \times LC$$

$$= 3.75 \text{ mm} + 10 \times 0.015$$

$$= 3.90 \text{ mm}$$

$$TV = MV - (\text{zero error})$$

$$= 3.9 - (-0.3) = 4.20 \text{ mm}$$

5. Ans (2.25)

$$W = 2\omega_0 = \frac{2}{\sqrt{LC}}$$

$$\text{and liner } XL = WL = 2\sqrt{\frac{L}{C}};$$

$$X_C = \frac{1}{WC} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

$$P = VI \cos \phi$$

$$P = \frac{V^2 R}{R^2 + (X_L - X_C)^2}$$

Energy supplied

$$E = PT = \frac{\pi R \sqrt{LC} V^2}{R^2 + 2.25 \left(\frac{L}{C} \right)}$$

6. Ans (20.00)

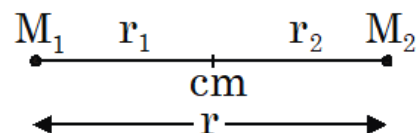
$$r_1 = \frac{M_2 r}{M_1 + M_2}$$

$$r_2 = \frac{M_1 r}{M_1 + M_2}$$

$$\frac{GM_1 M_2}{r^2} = M_2 \omega^2 r_2$$

$$\omega = \sqrt{\frac{GM_1}{r^2 r_2}}$$

$$T = \frac{2\pi \sqrt{r^2 r_2}}{\sqrt{GM_1}} = \frac{2\pi \sqrt{r^3}}{\sqrt{G(M_1 + M_2)}}$$

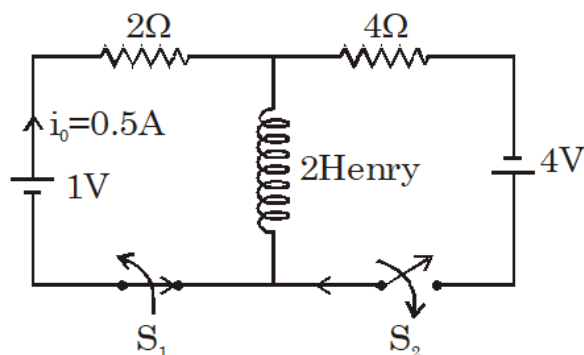


$$r = \sqrt[3]{G(M_1 + M_2) \left(\frac{T}{2\pi} \right)^2}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (3)



$$i_0 = 0.5 \text{ A}$$

When; S_2 is closed;

$$-L \frac{di}{dt} - 4i + 4 = 0$$

$$-2 \frac{di}{dt} - 4i_0 + 4 = 0$$

$$-2 \frac{di}{dt} + 4 \times 0.5 + 4 = 0 \quad [\because i_0 = -0.5 \text{ A}]$$

$$\frac{di}{dt} = 3$$

2. Ans (6)

At half power frequencies,

$$Z = R\sqrt{2} \Rightarrow |X_L - X_C| = R$$

$$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2 \Rightarrow \omega^2 L^2 + \frac{1}{\omega^2 C^2} - \frac{2L}{C} = R^2$$

$$\omega^4 L^2 C^2 - \omega^2 \left(\frac{2L}{C} + R^2\right) C^2 + 1 = 0$$

The product of roots is

$$\omega_1^2 \omega_2^2 = \frac{1}{L^2 C^2} = \omega_1 \omega_2 = \frac{1}{LC} = \omega_0^2$$

$$\Rightarrow \omega_0 = \sqrt{\omega_1 \omega_2} = 60 \text{ rad/s} \Rightarrow \omega_0/10 = 6$$

3. Ans (6)

$$1 \ell = 10^3 (\text{cc})^3$$

$$1 \ell = 10^{-3} \text{ m}^3$$

$$\text{Rate of water flowing} = \frac{1000}{7} \ell / \text{min}$$

$$= \frac{1}{7} \text{ m}^3 / \text{min} \{ \rho = 1000 \text{ kg/m}^3 \}$$

$$\text{Rate of water flow} = \frac{1000}{7} \text{ kg/min}$$

$$\text{Power of source} = P \times 10^6 \text{ watt/sec}$$

Let for a 60 sec power supply

$$\text{Energy of source} = P \times 10^6 \times 60$$

using energy conservation

$$P \times 10^6 \times 60$$

$$= \frac{1000}{7} \times 1 \times 4200 (60) + \frac{1000}{7} \times 1 \times 2268 \times 10^3$$

$$P \times 60 = \frac{42 \times 6}{7} + \frac{2268}{7} = 36 + 324$$

$$P \times 60 = 360$$

$$P = 6$$

4. Ans (4)

$$r_n = \frac{E_0 h^2 n^2}{\pi \mu e^2 Z}, \text{ where } \mu \text{ is the reduced mass}$$

$$\Rightarrow r \propto \frac{1}{\mu}$$

$$\text{Also ionisation energy } E = \frac{\mu Z^2 e^4}{8 \epsilon_0^2 h^2 n^2}$$

$$\Rightarrow E \propto \mu$$

For H-atom, $\mu = m_e$ and for positronium,

$$\mu = m_e/2$$

$$\Rightarrow \frac{r E_0}{r_0 E} = 4$$

5. Ans (3)

Using mirror formula,

$$\frac{1}{x_i} + \frac{1}{x_o} = \frac{2}{x_c}, \text{ we have } x_o = \frac{x_i x_c}{2x_i - x_c}$$

Also,

$$\frac{y_i}{y_o} = m = \frac{\frac{x_c}{2}}{\frac{x_c}{2} - x_o} = \frac{\frac{x_c}{2} - x_i}{\frac{x_c}{2}} = \frac{x_c - 2x_i}{x_c}$$

$$\Rightarrow y_o = \frac{y_i \cdot x_c}{x_c - 2x_i}$$

Now,

$$y_o = m x_o + C \Rightarrow \frac{y_i \cdot x_c}{x_c - 2x_i} = m \cdot \frac{x_i \cdot x_c}{2x_i - x_c} + C$$

$$\Rightarrow -y_i x_c = +m x_i \cdot x_c + 2x_i \cdot C - C x_c$$

$$\Rightarrow y_i = -x_i \left[m + \frac{2C}{x_c} \right] + C \Rightarrow m' = - \left[m + \frac{2C}{x_c} \right]$$

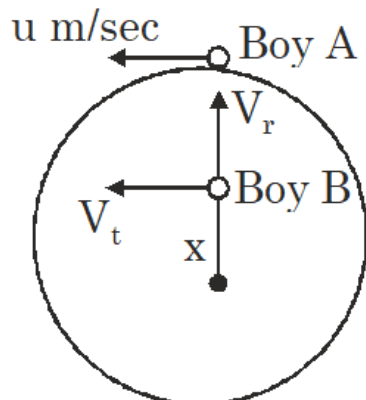
$$\text{Now, } \Rightarrow m = -\frac{4}{5}, C = 5, x_c = -50$$

$$\Rightarrow m' = - \left[-\frac{4}{5} + \frac{2 \times 5}{-50} \right] = 1$$

6. Ans (1)

at any time t

$$\omega_A = \omega_B = \omega/R$$



$$v_t = \omega B X$$

$$v_t = \frac{u}{R} x$$

$$v_b = u$$

$$\sqrt{V_t^2 + V_r^2} = u \quad V_r = \frac{dx}{dt}$$

$$\left(\frac{u}{R}\right)^2 x^2 + \left(\frac{dx}{dt}\right)^2 = u^2$$

$$\left(\frac{dx}{dt}\right)^2 = u^2 \left[1 - \frac{x^2}{R^2}\right]$$

$$\int_0^R \frac{dx}{\sqrt{1 - \frac{x^2}{R^2}}} = u \int_0^t dt$$

$$R \left[\sin^{-1} \frac{x}{R} \right]_0^R = ut$$

$$\frac{28 \times \frac{22}{7}}{2} = ut$$

$$t = 11 \text{ sec.}$$

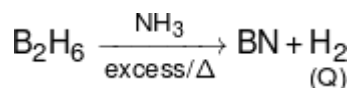
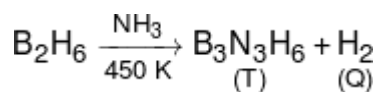
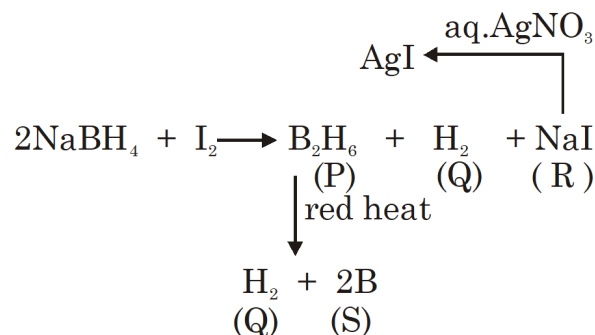
$$10 + x = 11$$

$$x = 1$$

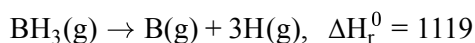
PART-2 : CHEMISTRY

SECTION-I

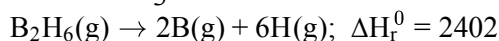
1. Ans (B,C,D)



BN is having graphite structure.



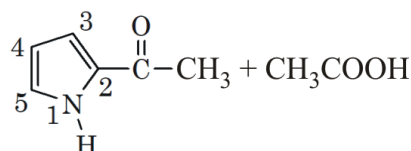
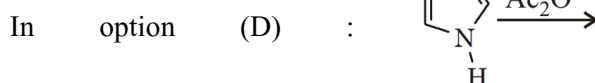
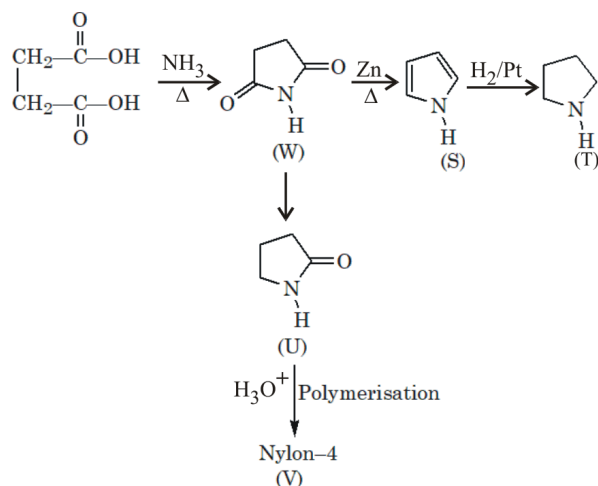
$$\text{BE}_{\text{B-H}} = \frac{1119}{3} = 373$$



$$\text{BE}_{3\text{c-}2\text{e}^- \text{ bond}} = \frac{2402 - 373 \times 4}{2} = 455 \text{ kJ/mol}$$

2. Ans (A,B)

'R' is least acidic.



3. **Ans (A,C)**

(A) is correct because

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_1 = T, \quad T_2 = \frac{T}{2}$$

$$V_1 = V \quad \text{and} \quad V_2 = 5.66 V$$

$$T V^{\gamma-1} = \frac{T}{2} (5.66 V)^{\gamma-1}$$

$$2 = (5.66)^{\gamma-1}$$

Taking logarithm on both the sides we get

$$\log 2 = (\gamma - 1) (\log 5.66)$$

$$0.3010 = (\gamma - 1) (0.7528)$$

$$\gamma - 1 = 0.3998$$

$$\gamma = 1.3998$$

$$\gamma = 1.4$$

$$\frac{C_p}{C_v} = \gamma = 1.4$$

So the gas is diatomic.

(C) is correct because work done

$$W = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = 1.25 P V$$

4. **Ans (B,D)**

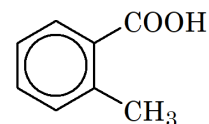
BBr_3 is having high lewis acid nature.

P_4O_{10} has 6 POP bonds.

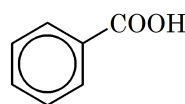
$ClOF_3$ & XeO_2F_2 are isostructural.

5. **Ans (B,C)**

Phenol does not give $NaHCO_3$ test.



is more acidic than



due to ortho-effect.

6. **Ans (A,B,C)**

$$K = A e^{-E_a/RT}$$

$$\frac{dK}{dT} = \frac{KE}{RT^2}$$

$$\frac{1}{K} \frac{dK}{dT} \Rightarrow \frac{E_a}{RT^2} \Rightarrow \sigma$$

$$E_a = \sigma RT^2 = \frac{1.25 \times 10^6}{T^3} RT^2$$

$$E_a \Rightarrow \frac{1.25 \times 10^6 R}{T}$$

(A)

$$T = 250 = \left(\frac{1.25 \times 10^6 \times 2}{250} \right) = 10^4 \text{ Cal/mol} = 10 \text{ K cal/mol}$$

(B)

$$T = 2000 \text{ K} = \frac{1.25 \times 10^6 \times 2}{2000} = 1.25 \times 10^3 \text{ Cal/mol}$$

$$\Rightarrow 1.25 \text{ K Cal/mol}$$

$$(C) \frac{dK}{dT} \propto \frac{1}{T^2}$$

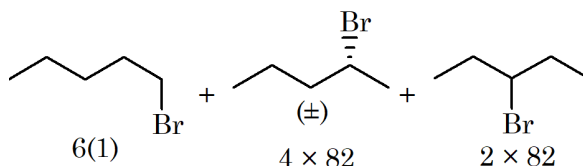
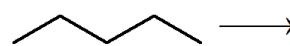
$$(D) \frac{dK}{dT} = \sigma K = \frac{1.25 \times 10^6}{(2000)^3} \times K$$

$$= 0.156 \times 10^{-3} K$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (32.93 to 33.00)**



% of R-2-bromopentane

$$\Rightarrow \frac{4 \times 82}{6 + (4 \times 82) + (2 \times 82)} \times 100 = \frac{65.86}{2} = 32.93$$

2. **Ans (51.20)**

$$\text{Reaction Quotient } Q_p = \frac{(P_{NH_3})^2}{(P_{H_2})^3 (P_{N_2})} \text{ at 1 hour}$$

$$Q_p = \frac{(200)^2}{(50)(2)^3} = 100$$

$$\Delta G = \Delta G^\circ + RT \ln Q_p \Rightarrow \Delta G = -(15.53 \times 2) + \frac{25}{3} \times \frac{300}{1000} \{ \ln 100 \}$$

$$\Delta G = -19.56 \text{ kJ/mol}$$

$$-19.56 \times 1000 = (-4.2 \times 1000) - (300) \Delta S$$

$$\Delta S = 51.2 \text{ JK}^{-1}$$

3. **Ans (14.00)**

In CrO_4^{2-} , number of Cr-O bonds = 4

In $\text{Cr}_2\text{O}_7^{2-}$, number of Cr-O bonds = 8

In CrO_2Cl_2 , number of Cr-O bonds = 2

4. **Ans (3.47 or 3.48)**

PCl_5 and PBr_5 both exist in ionic form in solid state, both give H_3PO_4 and HX on hydrolysis.

Bromine is liberated from NaBr on reaction with conc H_2SO_4 (this is not possible with NaCl).

Hence, a = 5, b = 3, c = 4 and atomic weight of

Bromine is 80 amu. Hence 80/23 value is 3.48.

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (5)**

Cell becomes

$$\begin{aligned}
 & \text{Pt, H}_2 \left(1 \text{ atm} \right) \left| \frac{\text{H}^+}{\sqrt{K_a C_1}} \right| \left| \frac{\text{H}^+}{\sqrt{K_b C_2}} \right| \text{Pt, H}_2 \left(1 \text{ atm} \right) \\
 & E_{\text{Cell}} = 0 - \frac{RT}{nF} \log \frac{[\text{H}^+]_{\text{left cell}}}{[\text{H}^+]_{\text{right cell}}} \\
 & = -0.0625 \log_{10} \frac{(K_a C_1)^{1/2}}{(K_b C_2)^{-1/2} \cdot K_w} \\
 & = -0.0625 \log \frac{\sqrt{10^{-5} \times 10^{-5} \times 0.1 \times 0.1}}{10^{-14}}
 \end{aligned}$$

$$= -0.0625 \times 8$$

$$E_{\text{Cell}} = -5 \times 10^{-1}$$

2. **Ans (4)**

(i), (ii), (iii), (vii) are the correct combinations

Magnetite Fe_3O_4

Gibbsite $\text{Al}(\text{OH})_3$

Copper pyrite CuFeS_2

Galena PbS

Cerussite PbCO_3

Lunar caustic AgNO_3

Dolomite $\text{MgCO}_3 \cdot \text{CaCO}_3$

3. **Ans (9)**

Theory based.

4. **Ans (2)**

$$\begin{cases} Z = \frac{N}{V} \cdot \sigma \cdot \sqrt{2} u_{\text{avg}} \\ \lambda = \frac{k_B}{\sqrt{2} \sigma} \left(\frac{T}{P} \right) \end{cases}$$

N = number of gas molecules

σ = collision area (depends on molecular size only)

$$u_{\text{avg}} = \text{average velocity of gas molecules} = \sqrt{\frac{8RT}{\pi M}}$$

Hence correct statements are (i) and (vii).

6. **Ans (2)**

1, 3, are correct.

PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,C)

$$I = \int_1^3 \frac{[x^2]}{[(x-4)^2] + [x^2]} dx \quad \dots(1)$$

$$I = \int_1^3 \frac{[(4-x)^2]}{[x^2] + [(4-x)^2]} dx \quad \dots(2)$$

By Adding (1) & (2)

$$\therefore 2I = \int_1^3 \frac{[x^2] + [(x-4)^2]}{[(x-4)^2] + [x^2]} dx = 2 \Rightarrow I = 1$$

Also,

$$\begin{aligned} \int_0^2 \left([x] - \left[\frac{x}{2} \right] \right) dx &= \int_0^1 [x] dx + \int_1^2 [x] dx - \int_0^2 \left[\frac{x}{2} \right] dx \\ &= \int_0^1 0 \cdot dx + \int_1^2 1 \cdot dx - \int_0^2 0 \cdot dx \\ &= 0 + 1 - 0 = 1 \end{aligned}$$

2. Ans (A,B,C,D)

Given equation

$$\int dy = \int e^x \left(\ln x + \frac{1}{x} \right) dx \Rightarrow y = e^x \cdot \ln x + c$$

$\therefore$ the curve passes through (1, 0)

$$\therefore 0 = e^x \ln 1 + c \Rightarrow c = 0$$

$$y = e^x \ln x \Rightarrow f(x) = e^x \cdot \ln x$$

$$(A) f(x) > 0 \forall x \in (1, \infty)$$

$$(B) \int e^{-x} \cdot e^x \ln x \, dx = \int \ln x \, dx = x \ln x - x + c$$

$$(C) \lim_{x \rightarrow 1} \frac{e^x \cdot \ln x}{x-1} = \lim_{h \rightarrow 0} \frac{e^{1+h} (\ln(1+h))}{h} = e$$

(Put $x = 1 + h$)

$$(D) \lim_{x \rightarrow 1} (1 + e^x \ln x)^{\frac{1}{x-1}} = \lim_{h \rightarrow 0} (1 + e^{1+h} \ln(1+h))^{\frac{1}{1+h-1}}$$

$$= 1^\infty \text{ form (Put } x = 1 + h)$$

$$= e^{\lim_{h \rightarrow 0} \frac{1}{h} (1 + e^{1+h} \ln(1+h) - 1)}$$

$$= e^{\lim_{h \rightarrow 0} \frac{e^{1+h} \ln(1+h)}{h}} = e^e$$

3. Ans (B,C)

$$\text{given } \frac{\vec{a} \cdot \vec{r}}{|\vec{a}|} = \frac{10}{\sqrt{3}}$$

$$\Rightarrow x + y + z = 10, x, y, z \in \mathbb{N}$$

$$\text{total number of positions} = {}^9C_2 = 36$$

(A) Number of natural solutions of

$$x + y + z = 7 \text{ is } {}^6C_2 = 15$$

$$(B) 6 \times 6 = 36$$

$$(C) \because 1800 = 2^3 \times 3^2 \times 5^2$$

$$\text{number of divisors} = (3+1)(2+1)(2+1) = 36$$

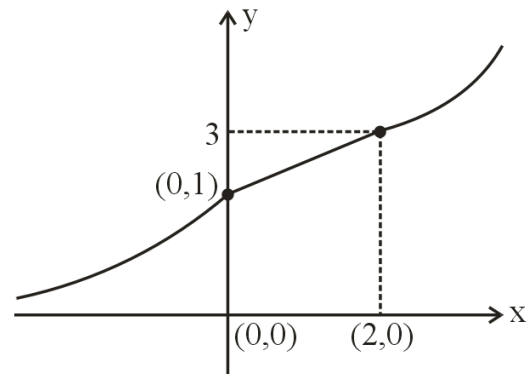
(D) Number of cyphers in $120!$ = exponent of 5 in $(120)!$

$$= \left[\frac{120}{5} \right] + \left[\frac{120}{25} \right] + \left[\frac{120}{125} \right] = 28$$

(where $[.]$ greatest integer function)

4. Ans (A,B,C)

$$f(0-h) = \lim_{h \rightarrow 0} 3^{-h} = 3^0 = 1$$



$$f(0+h) = \lambda$$

$f(x)$ is a continuous function and $\lambda = 1$

Range of $f(x)$ is $(0, \infty)$

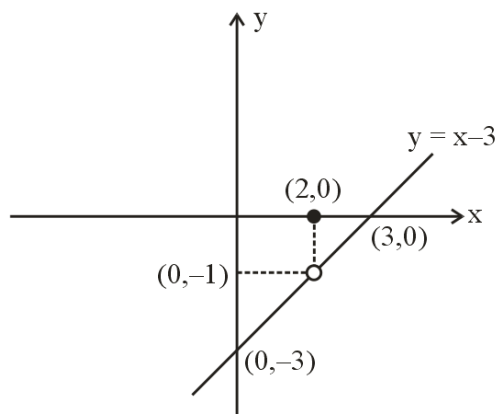
$f(x) = 1$ has one solution

$f(x) = |f(x)|$ is always possible

because $f(x) > 0$ for all $x \in \mathbb{R}$

5. Ans (A,B,C)

$$f(x)(x-2) = x^2 - 5x + 6 \therefore 'f' \text{ is continuous}$$



$$f(x) = \frac{x^2 - 5x + 6}{x - 2} = \frac{(x-3)(x-2)}{(x-2)} = x - 3,$$

for $x \neq 2$

$$f(x) = \begin{cases} x-3, & x \neq 2 \\ 0, & x = 2 \end{cases}$$

6. Ans (A,D)

$$f'(x) = -e^{-ax} (ax^2 + (a^2 - 2)x + ab - a)$$

$$\therefore f(-2) = f(1) = 0$$

$$\Rightarrow -\frac{(a^2 - 2)}{a} = -1 \text{ \& } \frac{ab - a}{a} = -2$$

$$\Rightarrow a = 2 \text{ \& } b = -1$$

$$f'(x) = -e^{-2x} (2x^2 + 2x - 4)$$

$$f'(x) = -2e^{-2x} (x^2 + x - 2)$$

$$f''(x) = 2e^{-2x} (2x^2 - 5)$$

$$f''(-2) = 6e^4 > 0$$

$$f''(1) = -6e^{-2} < 0$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.08)

$$k = \lim_{x \rightarrow 0} \frac{e^{-\frac{x^2}{2}} - \cos x}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\left(1 - \frac{x^2}{2} + \frac{x^4}{8} - \dots\right) - \left(1 - \frac{x^2}{2} + \frac{x^4}{4!} - \dots\right)}{x^4}$$

$$k = \frac{1}{8} - \frac{1}{24} = \frac{1}{12}$$

2. Ans (0.75)

E : Rose bush has withered.

A : Gardner did not water the rose bush

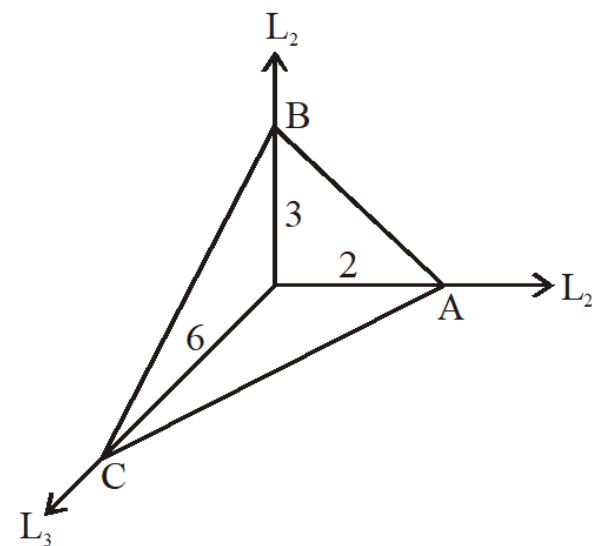
$$P(A) = \frac{2}{3}$$

$$B : \text{Gardner watered the rose bush } P(B) = \frac{1}{3}$$

$$P\left(\frac{E}{A}\right) = \frac{3}{4}; P\left(\frac{E}{B}\right) = \frac{1}{2}$$

$$P\left(\frac{A}{E}\right) = \frac{P(A) \cdot P\left(\frac{E}{A}\right)}{P(A) \cdot P\left(\frac{E}{A}\right) + P(B) \cdot P\left(\frac{E}{B}\right)} = \frac{\frac{2}{3} \cdot \frac{3}{4}}{\frac{2}{3} \cdot \frac{3}{4} + \frac{1}{3} \cdot \frac{1}{2}} = \frac{3}{4} = 0.75$$

3. Ans (1.60)



$$V = \frac{1}{3} (\text{Area of Base}) \times \text{Height}$$

$$6 = \frac{1}{3} \sqrt{\left(\frac{1}{2} \times 2 \times 3\right)^2 + \left(\frac{1}{2} \times 3 \times 6\right)^2 + \left(\frac{1}{2} \times 2 \times 6\right)^2} \times d$$

$$d = \frac{18}{\sqrt{126}} = \frac{6}{\sqrt{14}} \text{ units.}$$

4. Ans (1.33)

$$I = \int (f(x) - 2g(x)) dx = \int \frac{e^{3x} - 2e^x}{e^{4x} + 8e^{2x} + 4} dx$$

$$\text{Let } e^x = t$$

$$I = \int \frac{1 - \frac{2}{t^2}}{t^2 + 8 + \frac{4}{t^2}} dt = \frac{1}{2} \tan^{-1} \left(\frac{e^x + 2e^{-x}}{2} \right) + c$$

$$\text{Hence, } h(x) = \frac{1}{2} \tan^{-1} \left(\frac{e^x + 2e^{-x}}{2} \right)$$

$$h(0) = \frac{1}{2} \tan^{-1} \left(\frac{3}{2} \right)$$

$$\therefore a = 2, b = 3, c = 2.$$

5. **Ans (0.83)**

$$\text{RHS} = 2y^2 - 8y + 21 = 2(y-2)^2 + 13 \geq 13$$

$$\text{LHS} = 12\sin x + 5\cos x \leq 13$$

$$\therefore \text{RHS} = \text{LHS}$$

$$\Rightarrow 12\sin x + 5\cos x = 2y^2 - 8y + 21 = 13$$

$$\text{Now } 2(y-2)^2 + 13 = 13$$

$$\Rightarrow y = 2$$

$$\text{Similarly } 12\sin x + 5\cos x = 13$$

$$\Rightarrow \frac{12}{13}\sin x + \frac{5}{13}\cos x = 1$$

$$\text{or } \sin\left(x + \tan^{-1}\frac{5}{12}\right) = 1$$

$$\text{For least positive value of } \frac{x}{y}$$

$$\therefore \frac{x}{y} = \frac{\frac{\pi}{2} - \tan^{-1}\frac{5}{12}}{2} = \frac{1}{2}\cot^{-1}\left(\frac{5}{12}\right)$$

6. **Ans (0.66 or 0.67)**

$$\text{Line AB } \frac{x - \sqrt{3}}{\cos 60^\circ} = \frac{y}{\sin 60^\circ} = r$$

$$x = \sqrt{3} + \frac{r}{2}, y = \frac{r\sqrt{3}}{2}$$

$$\therefore \text{Point } \left(\sqrt{3} + \frac{r}{2}, \frac{r\sqrt{3}}{2}\right) \text{ lies on } 2y^2 = 2x + 3$$

$$\therefore \frac{3r^2}{2} = 2\sqrt{3} + r + 3$$

$$\Rightarrow 3r^2 - 2r - (6 + 4\sqrt{3}) = 0$$

PA and -PB are roots

$$\therefore \text{PA} - \text{PB} = \frac{2}{3}$$

$$\text{PA} \cdot \text{PB} = \frac{6 + 4\sqrt{3}}{3}$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (1)**

$$\text{Let } S_1 = \alpha + \beta + \gamma \text{ and } S_2 = \alpha\beta + \beta\gamma + \gamma\alpha \text{ \&}$$

$$S_3 = \alpha\beta\gamma$$

$$\therefore S_1 = 6 \text{ \& } (\alpha + 1)(\beta + 1)(\gamma + 1) = 33$$

$$\Rightarrow S_3 + S_1 + S_2 = 32$$

$$\Rightarrow S_2 + S_3 = 26 \quad \dots(1)$$

$$\text{Also, } \alpha^3 + \beta^3 + \gamma^3 = 87$$

$$\Rightarrow S_1(S_1^2 - 3S_2) + 3S_3 = 87$$

$$\Rightarrow 216 - 18S_2 + S_3 = 87$$

$$\Rightarrow 18S_2 - 3S_3 = 129$$

$$\Rightarrow 6S_2 - S_3 = 43 \quad \dots(2)$$

Adding (1) & (2)

$$\Rightarrow 7S_2 = 69 \Rightarrow S_2 = \frac{69}{7}$$

$$\& S_3 = \frac{113}{7}$$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{S_2}{S_3} = \frac{69}{113}$$

2. **Ans (7)**

$$|z_n| = \frac{13}{2^n}$$

$$\therefore z_n \text{ lies in part S} \Rightarrow |z_n| < 1$$

$$\Rightarrow n \geq 4$$

z_n also lies in IVth quadrant

$$\therefore \text{minimum value of } n \text{ is } 7$$

$n = 4$ then lie in Ist Quad

$n = 5$ then lie in IInd Quad

$n = 6$ then lie in IIIrd Quad

3. **Ans (0)**

$$\vec{c} = \vec{a} \times \vec{c} + \vec{a} \times \vec{b} \quad \dots\dots (i)$$

$$\Rightarrow \vec{a} \cdot \vec{c} = 0 \quad \dots(ii)$$

$$\text{Also } \vec{a} \times \vec{c} = \vec{a} \times (\vec{a} \times \vec{c}) + \vec{a} \times (\vec{a} \times \vec{b})$$

{Taking cross with $\vec{a}$ }

$$\vec{c} - \vec{a} \times \vec{b} = -\vec{a}^2 \vec{c} + (\vec{a} \cdot \vec{b}) \vec{a} - \vec{a}^2 \vec{b}$$

$$(1 + \vec{a}^2) \vec{c} + \vec{b} \times \vec{a} + (\vec{a} \times \vec{b}) \times \vec{a} = 0$$

4. **Ans (2)**

$$\log^2 y + 2 \left(2^x + \frac{1}{2^x} \right) \log y + 2 \left(2^{2x} + \frac{1}{2^{2x}} \right) = 0$$

$$\left(\log y + 2^x + \frac{1}{2^x} \right)^2 + 2 \left(2^{2x} + \frac{1}{2^{2x}} \right) = 2^{2x} + \frac{1}{2^{2x}} + 2$$

$$\underbrace{\left(\log y + 2^x + \frac{1}{2^x} \right)^2}_{\geq 0} = - \underbrace{\left(2^{2x} + \frac{1}{2^{2x}} \right) + 2}_{\leq 0}$$

$$\Rightarrow x = 0; y = e^{-2}$$

 5. **Ans (0)**

$$M = \begin{bmatrix} \sqrt{3} & 1 & 0 \\ 1 & -\sqrt{3} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\text{As, } MM^T = 4I$$

$$\Rightarrow M^T = 4M^{-1}$$

$$\Rightarrow M^T = 4 \left(\frac{\text{adj}M}{|M|} \right)$$

$$\Rightarrow 2M^T + \text{adj}.M = \mathbf{O}$$

$$\Rightarrow |2M^T + \text{adj}.M| = 0$$

 6. **Ans (9)**

$$P = {}^nC_6 \cdot \left(3^{\frac{1}{3}} \right)^{n-6} \cdot \left(4^{-\frac{1}{3}} \right)^6$$

$$Q = {}^nC_{n-6} \cdot \left(3^{\frac{1}{3}} \right)^6 \cdot \left(4^{-\frac{1}{3}} \right)^{n-6}$$

$$\therefore \frac{Q}{P} = 12 \Rightarrow (12)^{\frac{n-6}{3}} = (12)^1$$

$$\Rightarrow \frac{n-6}{3} = 1 \Rightarrow n = 9$$